WEEK-2

Program:

```sql
-- Step 1: Create Database
CREATE DATABASE Donation_db;
USE Donation_db;
-- Step 2: Create Books Table
CREATE TABLE Books(
    book_id INT PRIMARY KEY,
    title VARCHAR(100) NOT NULL,
    isbn VARCHAR(20) UNIQUE,
    published_year INT CHECK(published_year < 2027)
);
-- Step 3: Insert Records into Books Table
INSERT INTO Books VALUES
(1,'The Great Gatsby','9780743273565',1925),
(2,'To Kill a Mockingbird','9780061120084',1960),
(3,'1984','9780451524935',1949);
```

Output:

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	13:30:09	CREATE DATABASE Donation_db	1 row(s) affected	0.016 sec
2	13:30:11	USE Donation_db	0 row(s) affected	0.000 sec
3	13:30:13	CREATE TABLE Books(book_id INT PRIMARY KEY, title VAR...	0 row(s) affected	0.047 sec
4	13:30:15	INSERT INTO Books VALUES (1,'The Great Gatsby','978074327356...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec

Program:

```sql
-- Step 4: Display Books Table
SELECT * FROM Books;
-- Step 5: Create Members Table
CREATE TABLE Members(
    member_id INT PRIMARY KEY,
    full_name VARCHAR(100),
    email VARCHAR(100) UNIQUE
);
-- Step 6: Insert Records into Members Table
INSERT INTO Members VALUES
(101,'Tejaswini','iliharshika@gmail.com'),
(102,'Bhanu Priya','bhanupriya@gmail.com'),
(103,'Sahasra','sahasra@gmail.com');
```

Output:

5	13:30:49	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
6	13:31:01	CREATE TABLE Members(member_id INT PRIMARY KEY, f...	0 row(s) affected	0.078 sec
7	13:31:05	INSERT INTO Members VALUES (101,'Tejaswini','iliharshika@gm...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec

Result Grid | Filter Rows: | Edit:

book_id	title	isbn	published_year
1	The Great Gatsby	9780743273565	1925
2	To Kill a Mockingbird	9780061120084	1960
3	1984	9780451524935	1949
NULL	NULL	NULL	NULL

Program:

```sql
SELECT * FROM Members;
```

Output:

member_id	full_name	email
101	Tejaswini	iliharshika@gmail.com
102	Bhanu Priya	bhanupriya@gmail.com
103	Sahasra	sahasra@gmail.com
NULL	NULL	NULL

Members 4 ×

Output

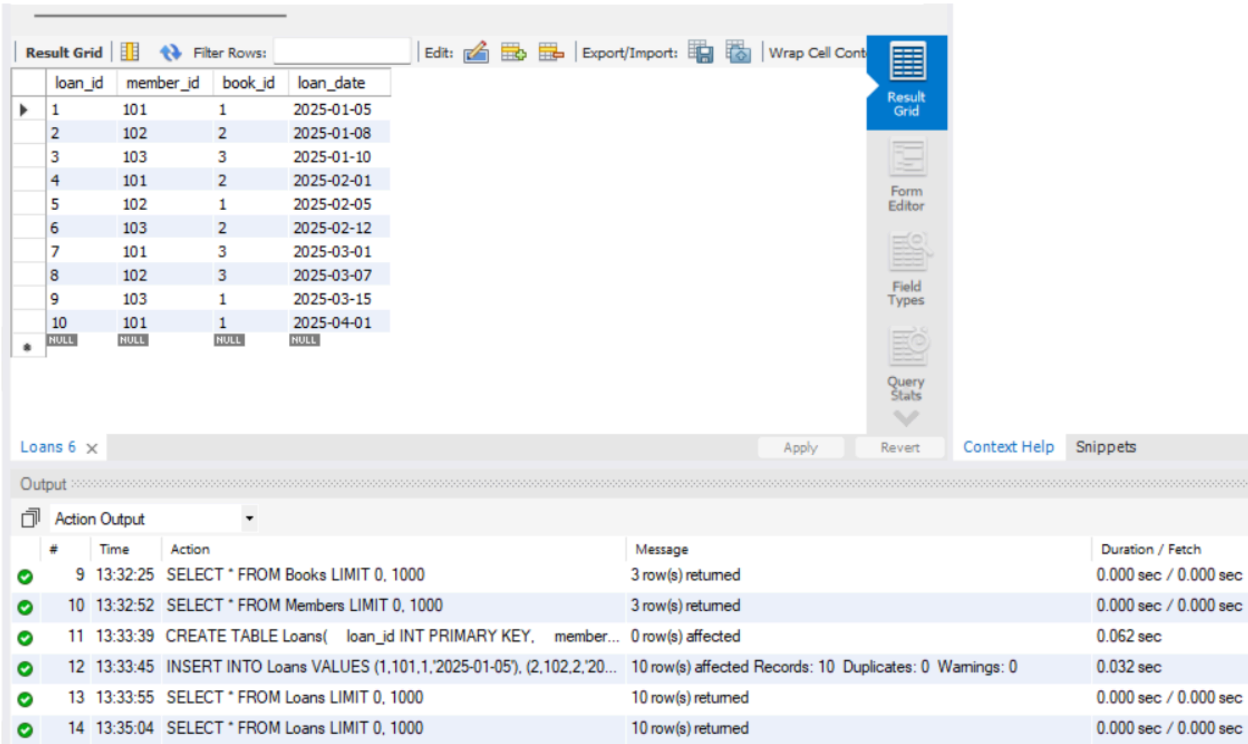
Action Output

#	Time	Action	Message	Duration / Fetch
5	13:30:49	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
6	13:31:01	CREATE TABLE Members(member_id INT PRIMARY KEY, f...	0 row(s) affected	0.078 sec
7	13:31:05	INSERT INTO Members VALUES (101,'Tejaswini','iliharshika@gm...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
8	13:31:08	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
9	13:32:25	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
10	13:32:52	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec

Program:

```sql
-- Step 8: Create Loans Table
CREATE TABLE Loans(
    loan_id INT PRIMARY KEY,
    member_id INT,
    book_id INT,
    loan_date DATE,
    FOREIGN KEY(member_id) REFERENCES Members(member_id),
    FOREIGN KEY(book_id) REFERENCES Books(book_id)
);
-- Step 9: Insert Records into Loans Table
INSERT INTO Loans VALUES
(1,101,1,'2025-01-05'),
(2,102,2,'2025-01-08'),
(3,103,3,'2025-01-10'),
(4,101,2,'2025-02-01'),
(5,102,1,'2025-02-05'),
(6,103,2,'2025-02-12'),
(7,101,3,'2025-03-01'),
(8,102,3,'2025-03-07'),
(9,103,1,'2025-03-15'),
(10,101,1,'2025-04-01');
-- Step 10: Display Loans Table
SELECT * FROM Loans;
```

Output:



loan_id	member_id	book_id	loan_date
1	101	1	2025-01-05
2	102	2	2025-01-08
3	103	3	2025-01-10
4	101	2	2025-02-01
5	102	1	2025-02-05
6	103	2	2025-02-12
7	101	3	2025-03-01
8	102	3	2025-03-07
9	103	1	2025-03-15
10	101	1	2025-04-01
NULL	NULL	NULL	NULL

#	Time	Action	Message	Duration / Fetch
9	13:32:25	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
10	13:32:52	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
11	13:33:39	CREATE TABLE Loans(loan_id INT PRIMARY KEY, member...	0 row(s) affected	0.062 sec
12	13:33:45	INSERT INTO Loans VALUES (1,101,1,'2025-01-05'), (2,102,2,'20...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.032 sec
13	13:33:55	SELECT * FROM Loans LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
14	13:35:04	SELECT * FROM Loans LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec

Program:

```
SELECT
    m.full_name AS Member_Name,
    b.title AS Book_Title
FROM Loans l
INNER JOIN Members m
ON l.member_id = m.member_id
INNER JOIN Books b
ON l.book_id = b.book_id;
```

Output:

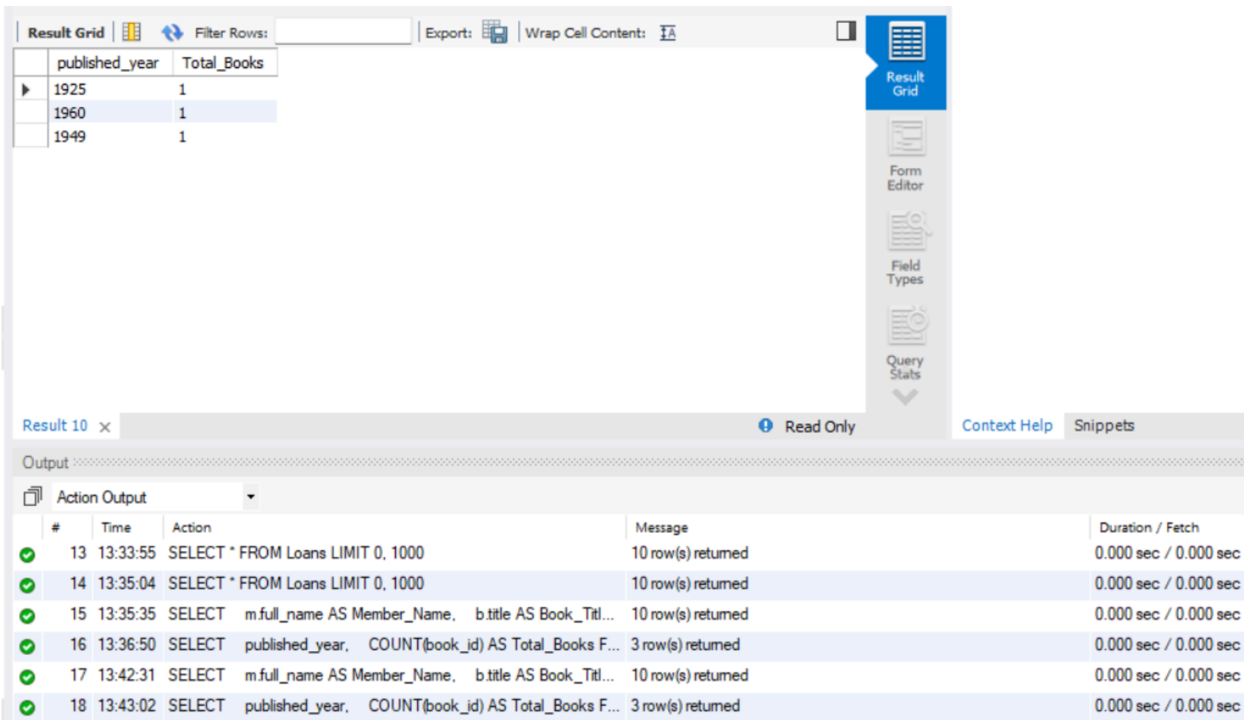
Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Member_Name	Book_Title
Tejaswini	The Great Gatsby
Tejaswini	To Kill a Mockingbird
Tejaswini	1984
Tejaswini	The Great Gatsby
Bhanu Priya	To Kill a Mockingbird
Bhanu Priya	The Great Gatsby
Bhanu Priya	1984
Sahasra	1984
Sahasra	To Kill a Mockingbird
Sahasra	The Great Gatsby

Program:

```sql
SELECT
    published_year,
    COUNT(book_id) AS Total_Books
FROM Books
GROUP BY published_year;
```

Output:



Program:

```sql
-- Step 13: Create Donation History Table
CREATE TABLE Donation_History(
    donation_id INT PRIMARY KEY,
    book_id INT,
    donor_name VARCHAR(100),
    donation_date DATE,
    FOREIGN KEY(book_id) REFERENCES Books(book_id)
);
-- Step 14: Start Transaction
START TRANSACTION;
-- Step 15: Insert New Book into Books Table
INSERT INTO Books VALUES
(4,'Animal Farm','9780451526342',1945);
-- Step 16: Insert Donation Record and Commit Transaction
INSERT INTO Donation_History VALUES
(1,4,'Raj Kumar',CURDATE());

COMMIT;
```

Output:

✔	19	13:44:08	CREATE TABLE Donation_History(donation_id INT PRIMARY ...	0 row(s) affected	0.109 sec
✔	20	13:44:13	START TRANSACTION	0 row(s) affected	0.000 sec
✔	21	13:44:16	INSERT INTO Books VALUES (4,'Animal Farm','9780451526342',...	1 row(s) affected	0.032 sec
✔	22	13:44:21	INSERT INTO Donation_History VALUES (1,4,'Raj Kumar',CURDA...	1 row(s) affected	0.031 sec
✔	23	13:44:24	COMMIT	0 row(s) affected	0.016 sec

Program:

```
-- Step 17: Create Index on ISBN Column
CREATE INDEX idx_isbn
ON Books(isbn);
-- Step 18: Search Book Using ISBN
SELECT *
FROM Books
WHERE isbn='9780451524935';
```

Output: